A nonzero Szegő correction on the smooth circle

Disproof of Conjecture 00000009774

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Abstract

The strictly positive analytic symbol $f(\theta) = 5/4 + \cos\theta$ on the unit circle has Toeplitz determinants $D_n = \frac{4}{3}(1 - 4^{-(n+1)})$. Its logarithmic leading coefficient is zero, but its secondary term tends to $\log(4/3) > 0$. The circle has no corners, so the claimed disappearance of the smooth secondary term and the absence of a global correction are false. Lean verifies the actual Toeplitz matrices, their determinants for every size, and both limits.

The assertion being tested

The bilingual source states that the secondary term of logarithmic Toeplitz determinant asymptotics depends entirely on boundary corners, vanishing in the smooth case. We test this necessary smooth-case subclause in the classical circle setting of Szegő's theorem. The unit circle is a smooth boundary, with no corner angles. A universal claim for smooth domains must in particular hold in this case.

For a positive circle symbol f, write

$$\widehat{f}(k) = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-ik\theta}\,d\theta, \quad T_n(f) = (\widehat{f}(i-j))_{0\le i,j<n}, \quad D_n(f) = \det T_n(f).$$

These are the finite sections of the classical Toeplitz operator, compression of multiplication by f to the span of $1, e^{i\theta}, \dots, e^{i(n-1)\theta}$ in the circle Hardy space. The logarithmic leading coefficient is

$$L(f) = \frac{1}{2\pi}\int_0^{2\pi} \log f(\theta)\,d\theta.$$

The tested assertion requires $\log D_n(f) - nL(f) \to 0$ for smooth positive symbols on this smooth boundary. Our example has $L(f) = 0$, so changing the size convention from n to $n+1$ cannot change the secondary constant.

An analytic positive symbol and its matrix

Choose

$$f(\theta) = \frac{5}{4} + \cos\theta = \left|1 + \tfrac{1}{2}e^{i\theta}\right|^2.$$

It is real analytic and 2π-periodic, with $f \ge 1/4 > 0$. The elementary exponential formula for cosine gives

$$\widehat{f}(0) = \tfrac{5}{4}, \quad \widehat{f}(1) = \widehat{f}(-1) = \tfrac{1}{2}, \quad \widehat{f}(k) = 0 \quad (|k| > 1).$$

Consequently $T_n(f)$ is the actual tridiagonal Toeplitz matrix with diagonal $5/4$ and both adjacent diagonals $1/2$.

The power series for $\log(1+z)$, evaluated at $z = e^{i\theta}/2$, gives

$$\log f(\theta) = 2\,\mathrm{Re}\sum_{k\ge1} \frac{(-1)^{k+1}}{k\,2^k}e^{ik\theta}.$$

This series converges absolutely and uniformly, since $\sum_{k\geq 1} 1/(k2^k) < \infty$. Termwise integration is therefore valid. Every nonconstant exponential has zero circle mean, so $L(f) = 0$. This is also verified independently as the leading coefficient by the exact determinant asymptotics below.

Exact determinants and the nonzero correction

Theorem 1. *For every integer $n \geq 0$, with the empty determinant $D_0 = 1$,*

$$D_n(f) = \frac{4}{3}\left(1 - \left(\frac{1}{4}\right)^{n+1}\right) > 0.$$

Proof. For a tridiagonal Toeplitz matrix with diagonal d and adjacent diagonals a, expansion in the first row, followed by expansion of the off-diagonal minor in its first column, gives

$$D_{n+2} = dD_{n+1} - a^2 D_n.$$

For the present matrix this becomes $D_{n+2} = \frac{5}{4}D_{n+1} - \frac{1}{4}D_n$, with $D_0 = 1$ and $D_1 = 5/4$. The proposed expression has these initial values and satisfies the recurrence, since its two exponential bases are 1 and $1/4$. Induction proves the identity. Positivity follows from $0 < (1/4)^{n+1} < 1$. □

It follows immediately that

$$D_n(f) \longrightarrow \frac{4}{3}, \qquad \log D_n(f) \longrightarrow \log\frac{4}{3} > 0, \qquad \frac{\log D_n(f)}{n} \longrightarrow 0.$$

Thus the actual leading coefficient is zero, consistently with the circle integral, and

$$\log D_n(f) - nL(f) \longrightarrow \log\frac{4}{3} \neq 0.$$

Equivalently, the normalized determinant has a nonunit multiplicative factor $4/3$. There are no corners that could supply this factor. This refutes the smooth-vanishing assertion and hence the stated corner-only secondary law. No general form of the strong limit theorem is needed for the counterexample.

Formal verification and scope

Lean defines the coefficients of the general Laurent symbol $f_a(\theta) = 1+a^2+2a\cos\theta$ by $c_0 = 1+a^2$, $c_{\pm 1} = a$, with all other coefficients zero. It constructs the actual Mathlib matrix $(c_{i-j})_{0\leq i,j<n}$, proves its equality to the tridiagonal matrix, and proves the determinant recurrence by Mathlib's Laplace expansion. It then proves the exact determinant formula for every n, positivity, the logarithmic limit, and the normalized leading limit. The symbol's smoothness and positivity at $a = 1/2$ are also formalized.

The final proposition quantifies over the entire family f_a, with smoothness and positivity hypotheses. It asserts that every logarithmic leading limit L has a vanishing centered secondary term. This is a necessary universal subclause of the source's smooth-vanishing assertion. The theorem

`Conjecture9774.conjecture9774_false`

negates that proposition using $a = 1/2$ and $L = 0$. The contradiction uses uniqueness of limits and $\log(4/3) > 0$. The scalar formula in Lean is proved equal to actual matrix determinants; it does not substitute an unrelated scalar sequence for Toeplitz determinants.

The identification of the Laurent coefficients with Fourier integrals, the Hardy-space compression description, the smooth geometry of the unit circle, and the mean-log integral computation are supplied on paper. Lean does not implement general Toeplitz operators or a general Szegő theorem. Its normalized determinant limit independently proves the leading coefficient needed for the formal contradiction. Scalar extension of the real matrices to complex matrices preserves their determinants, so using real matrices does not change this example.

Source and reproduction

The unmodified bilingual source is supplied as `SOURCE.md`. Frozen provenance and its SHA-256 appear in the verification folder. The coordinating workflow found no prior submission for this conjecture. Lean is pinned to 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

Run in the supplied Lean folder:

```
lake env lean -DwarningAsError=true Main.lean
```

The printed axioms are only `propext`, `Classical.choice`, and `Quot.sound`. This draft was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.